

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering ASSESSMENT TOOL 1 – CLASS TEST (5 Questions)

Course Code:	ITA05	Course Title:	COMPUTER VISION
CO Assessed:	CO2– Manipulation (BL3)	Assessment:	Assessment Tool 3 – Class Test
Weightage:	20% of CIA	Total Marks:	25 (5 Questions × 5 Marks)
CO5 Statement:	Apply image enhancement and segmentation techniques for extracting meaningful regions from images.(BL3)		
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Section / Batch:	slot-D / 2024	Date:	11/8/26

Code of Conduct

I T.V.S.Jagadeesh / 192472146 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: T.S.J

Instructions

- This test contains 5 questions, each carrying 10 marks. Total: 50 Marks.
- No negative marking. Unanswered questions carry zero marks.
- No calculators or electronic devices permitted.
- Duration: 60 minutes.

Section / Topic	Focus	Q Nos	Marks
Image Enhancement in Spatial Domain	Improving image quality using spatial domain techniques	1	10
Histogram Processing Techniques	Enhancing image contrast through histogram equalization, histogram specification, and histogram stretching.	2	10
Spatial Filtering	Noise reduction and feature enhancement	3	10
Image Enhancement in Frequency Domain	Enhancing images using frequency domain transformations	4	10
Image Segmentation	Partitioning images into meaningful regions	5	10
GRAND TOTAL:			50 Marks

1. Explain the concept of image enhancement in the spatial domain. Discuss how basic gray-level transformations, histogram processing techniques, and spatial filtering can be used to improve the quality of the captured images. Justify the suitability of each technique for this scenario.
2. Differentiate between image smoothing and image sharpening. Explain the working principles of spatial smoothing filters and sharpening filters. Discuss how these techniques improve image quality and explain the situations in which each technique should be applied.
3. Explain the image enhancement process in the frequency domain. Describe the roles of low-pass filters and high-pass filters in image enhancement. Compare their effects on image quality and discuss suitable applications for each filtering technique.
4. Explain the concept of image segmentation. Compare point detection, line detection, edge detection, thresholding, and region-based segmentation with respect to their working principles, advantages, limitations, and practical applications. Identify the most appropriate technique(s) for this scenario and justify your answer.
5. Explain the importance of edge detection, edge linking, and boundary detection in image processing. Discuss the working principles of these techniques and describe how they contribute to accurate object recognition and defect detection. Illustrate your answer with suitable examples and diagrams wherever necessary.

Rubrics for Calculation

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning	Marks
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Q1. Image Enhancement in Spatial domain

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Introduction:-

→ Image enhancement in the spatial domain involves directly modifying pixel value to improve quality.

$$g(x, y) = T[f(x, y)]$$

↓ input image
O/p image

i) Gray level transformation:-

These are point processing techniques
formula: $s = 255 - r$

Types:-

- * Negative transformation
- * Log transformation
- * Power-law (gamma)

ii) Histogram processing

It represents intensity distribution

Technique

- * Histogram equalization
- * Histogram stretching

iii) Spatial filtering:-

Uses neighborhood pixels

Techniques:

- * Smooth filter
- * Sharpening filter

Q2. Image Smoothing vs Sharpening

Difference:-

Feature	Smoothing	Sharpening
Purpose	Reduce noise	Enhance edge
Technique	Gaussian, median etc	Sobel, Un-sharp etc
Effect	Blur	Highlight details

i) Smoothing filter

Replaces pixel with neighborhood average

$$g(x, y) = \frac{1}{mn} \sum f(i+x, i+y)$$

Effect:-

reduces noise

Blur image

ii) Sharpening filter

Enhancing the image through pixel

$$g(x, y) = f(x, y) - \nabla^2 f(x, y)$$

Technique:-

Effect:

→ Highlight the details

→ Clear edges

Q3. Image Enhancement process in frequency domain

→ It convert spatial domain (image) into Fourier transformation.

$$F(u,v) = \mathcal{F}(f(x,y))$$

$f(x,y)$ → input spatial image

$F(u,v)$ → Fourier transformation

Low-Pass filter:-

→ ~~Reduces~~ Removes high-frequency components

→ Smooth range

Effect:-

→ Reduce noise

→ Blurs edge

High-pass filter:-

→ Reduce low-frequency components

→ sharpening (enhance edge).

Effect:-

→ Sharpen image

→ Highlights details.

Q4 Image Segmentation

→ It divided ^(separate) the color, shape, and objects etc.

Point detection:-

- To detect ^{isolate} linear pixel structure
- Sensitive to noise

Line detection:-

- To detect linear structure
- Used in road detection

Edge detection:-

- To detect the boundaries
- Uses for gradients.

Threshold detection:-

- Separates based on intensity
- Simple but limited

Region - Based

- Groups similar pixels
- more accurate.

Q5. Edge detection, Linking, Boundary detection

Edge detection:-

→ Identifies intensity discontinuities

methods:-

* Sobel

* Canny

Edge linking:-

→ Connects edge pixel into meaningful structure

methods:-

* Hough transform

* Graph-based linking

~~Boundary detection:-~~

~~→ It defines the outline of the image for identification.~~

Contribution:-

Edge detection → Identity

Edge linking → ^{form} shape

Boundary → Object.

Application:-

→ Medical images

→ satellite images.